

URMIL JATIN CHANDARANA

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EDUCATION:

University of California, Davis

Master of Science in Computer Science

Sept 2025 - Present

GPA: 4.0 / 4.0

Relevant Coursework: ECS 265: Distributed Database Systems; ECS 235A: Computer and Information Security; ECS 252: Computer Networks; ECS 273: Visual Analytics;

K.J. Somaiya College of Engineering, Mumbai

Bachelor of Technology, Computer Engineering

Aug 2018 – Jun 2022

GPA: 9.4 / 10

Relevant Coursework: Advanced Algorithms: Design and Analysis; Operating Systems; Full Stack Development; Computer Network; Blockchain; Cloud Computing Emerging Technologies.

WORK EXPERIENCE:

Graduate Student Researcher

Davis, California

University of California – Davis | Project AgML, PAIBL Lab | Advisor: Prof. Mason Earles

Apr 2026 – Present

- Designed and built an **automated VLM dataset ingestion pipeline** leveraging an **open-source reasoning model** that analyzes raw datasets, **generates schema mappings**, and convert outputs to **Parquet format** for direct upload to **HuggingFace**.
- Developed a **dataset benchmarking framework** across structural quality, content difficulty, diversity, and annotation reliability, producing a **leaderboard** to guide selection of future agricultural VLM benchmark datasets.
- Building a **prompt-translation module** combining work on **OFAT and combinatorial search** over an LLM/VLM to auto-generate optimal per-dataset, per-model prompts, **improving zero-shot object detection accuracy** for agricultural VLMs.
- Engineered large-scale data processing workflows (**600K+ samples**) on **SLURM-managed HPC infrastructure**, resolving PyArrow/HuggingFace pipeline bottlenecks for efficient dataset conversion at scale.
- Collaborated with dataset authors to formalize data-sharing agreements and integrate external agricultural datasets into the AgML ecosystem.

Associate Consultant (Software Development Engineer)

Pune, India

Oracle Financial Services Software Limited

Aug 2022 – Jun 2025

- Engineered 10+ core banking modules** for the Oracle Banking Digital Experience (OBDX) platform, using **Spring Boot, Java 8, REST APIs/SOAP APIs & Microservices** for a major European Bank.
- Streamlined DevOps pipelines, reducing manual intervention by 30%** using tools including **Jenkins & Gradle** and implemented automation guidelines to enhance team efficiency.
- Orchestrated the migration** from a frequent traditional application setup to a **containerized environment** using **CI/CD pipelines & Docker**, reducing environment setup and deployment time by **85%** while **achieving 100% environment parity** across SDLC stages.
- Spearheaded the development and delivery of critical modules**, including a Login system via CAS & Bill Payment Functionality, ensuring seamless integration under tight deadlines alongside collaborating with clients.
- Led 7 product patch set upgrades**, ensuring meticulous planning and error-free execution and **acted as a technical SPC (Single point-of-contact)** for client stakeholders.
- Optimized team efficiency by 75%** through **10+ technical enablement sessions** and multiple requirement workshops; successfully mitigated requirement-related bugs and accelerated the adoption of new product features.

TECHNICAL SKILLS:

Programming Languages	Java, C++, JavaScript, Python, SQL, TypeScript, HTML
Libraries & Frameworks	Spring Boot, Node.js, Django, FastAPI, Flutter, Next.js, React.js, Tailwind CSS
ML / Data	PyTorch, HuggingFace, PyArrow, Ollama, vLLM, Model Quantization, Scikit-learn
Tools / Cloud	Bazel, MongoDB, DynamoDB, Redis, Docker, Kubernetes, Git, GitHub Actions, AWS

PROJECTS:

AgML-Benchmarking: An Agricultural Dataset Benchmarking Pipeline 🐍

Apr 2026 – Present

Python, PyTorch, HuggingFace, DINOv2, ResNet-18, FAISS, UMAP, scikit-learn, cleanlab, SLURM, Docusaurus, Scattergl, React, TypeScript

- Built a multi-phase ML dataset benchmarking pipeline** for agricultural image classification, evaluating **10+ quality metrics across 4 axes** (structural quality, content difficulty, diversity, annotation reliability) on HuggingFace datasets ranging from **500 to 60k images**, running on a university HPC cluster via SLURM.
- Implemented embedding-based dataset analysis using **DINOv2** with **FAISS** for near-duplicate detection and **UMAP for 2D/3D visualization**, surfacing cross-split contamination and near-duplicate rates — **flagged 27 cross-split leaks** in one dataset, directly inflating its reported test accuracy.
- Designed a **Dataset Cartography system** by fine-tuning **ResNet-18 over 30 epochs** to classify training samples as easy/ambiguous/hard, combined with **Cleanlab confident learning** (5-fold stratified CV on DINOv2 embeddings) for automated mislabeling detection — **identified a 40.8% estimated noise rate** in a grain disease dataset with **<40% classification accuracy** across all classes.
- Engineered a **composite 0-10 scoring framework** with anti-gaming constraints across 4 weighted axes, computed client-side so formula updates require no pipeline reruns, with **automated SLURM orchestration** for batch multi-dataset submission.

Localbite: Community-First Food Delivery Platform 🐍

Feb 2026 – May 2026

Python, FastAPI, TypeScript, SQL, Redis, Kafka, Docker, WebSocket, OAuth2, Tailwind CSS, Next.js, Cursor

- Architected a local-first food delivery ecosystem** to disrupt high-fee marketplaces, **reducing restaurant commission structures from 30% down to 8-12%** while keeping revenue within the Davis student community for **15,000+ potential local users**.
- Designed a Student-First phased dispatch engine** utilizing **Redis** for low-latency job queuing and **WebSocket** for real-time bidding; the system enforces a 4-minute exclusive window for student couriers before dynamically broadcasting to a wider pool.
- Engineered an asynchronous, service-oriented backend** designed for **event-driven scalability**, utilizing **Docker** to containerize microservices including a BaseFare calculation engine, payment settlement, and real-time notification service.